

KHANPUR, Pakistan

WMO: 417180

Lat: 28.650N

Lon: 70.683E

Elev: 289

StdP: 14.54

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.6	41.8	32.7	27.4	55.7	35.3	30.5	54.7	11.7	65.0	9.7	64.2	0.1	N/A	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.5	111.2	77.8	108.4	77.9	106.0	78.7	86.3	99.7	85.4	98.7	84.7	98.0	3.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.3	175.9	94.2	82.4	170.7	94.0	81.6	165.8	93.4	51.6	100.4	50.5	98.5	49.7	98.6	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.6	11.5	9.2	DB	35.7	115.6	2.7	2.2	33.7	117.2	32.2	118.5	30.7	119.7	28.8	121.3
			WB	35.2	88.4	3.1	0.8	33.0	88.9	31.2	89.4	29.5	89.8	27.3	90.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.0	56.1	62.6	73.0	85.4	93.6	95.1	93.4	90.5	87.6	80.8	69.3	59.1
	DBStd	14.38	4.13	5.38	6.24	5.73	4.11	3.31	3.09	2.40	2.80	4.27	4.40	4.54
	HDD50	6	5	0	0	0	0	0	0	0	0	0	0	1
	HDD65	585	277	101	10	0	0	0	0	0	0	0	9	188
	CDD50	10575	193	353	714	1062	1352	1354	1347	1255	1127	955	579	284
	CDD65	5680	0	34	259	612	887	904	882	790	677	490	138	7
	CDH74	81881	34	404	2629	8189	14007	14649	14024	11728	9236	5468	1364	149
	CDH80	52032	0	92	1198	5118	9699	10354	9579	7281	5276	2992	432	11
Wind	WSAvg	2.1	1.2	2.1	2.1	2.5	2.9	3.8	3.4	2.7	2.0	1.0	0.8	0.9
Precipitation	PrecAvg	3.5	0.2	0.1	0.2	0.1	0.2	0.2	1.1	1.1	0.4	0.0	0.0	0.2
	PrecMax	10.4	1.1	0.4	1.2	0.4	0.9	1.6	4.4	5.8	3.6	0.1	0.2	1.1
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.7	0.3	0.1	0.4	0.1	0.2	0.4	1.5	1.5	1.0	0.0	0.0	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.7	87.0	100.3	111.3	114.8	113.4	107.9	102.8	103.0	99.9	90.8	81.7
		MCWB	59.4	64.8	71.6	72.7	76.7	79.8	83.3	83.7	82.5	77.1	70.1	64.9
	2%	DB	73.9	82.7	95.1	107.4	112.1	110.3	105.2	101.3	100.5	98.0	88.2	77.9
		MCWB	58.3	64.2	70.3	71.1	77.2	80.3	84.1	83.5	81.8	76.7	69.8	62.1
	5%	DB	71.1	78.9	91.0	104.1	109.3	107.7	103.6	99.9	98.8	95.8	85.0	74.9
		MCWB	57.3	63.2	69.0	71.1	77.0	80.9	84.1	83.4	81.3	75.7	68.2	60.1
	10%	DB	68.1	75.5	86.8	100.5	106.4	105.3	101.8	98.3	96.9	93.5	81.8	71.8
		MCWB	55.6	61.2	67.6	71.2	76.8	81.6	84.0	83.2	80.4	74.3	66.2	58.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	69.3	75.5	78.5	84.9	87.0	87.4	86.3	85.8	81.5	73.5	66.3
		MCDB	72.5	80.5	94.2	100.4	104.0	102.3	100.9	98.2	98.5	95.3	87.1	78.8
	2%	WB	60.1	67.0	72.9	76.0	82.8	85.5	86.5	85.4	84.6	79.7	71.2	63.9
		MCDB	70.9	78.1	90.3	96.4	103.0	101.8	99.8	96.9	96.1	92.9	84.6	75.1
	5%	WB	58.5	64.7	70.8	74.5	81.4	84.7	85.6	84.7	83.4	78.1	69.8	61.9
		MCDB	69.0	76.5	86.4	95.6	101.5	101.2	98.8	96.1	94.5	91.1	83.0	72.5
	10%	WB	57.0	62.4	68.9	73.1	79.9	83.7	84.9	84.0	82.3	76.6	68.0	59.6
		MCDB	66.1	73.0	83.4	93.7	99.6	100.2	98.1	95.1	93.3	89.7	79.4	69.4
Mean Daily Temperature Range	5% DB	MDBR	24.0	25.0	25.0	27.7	26.1	21.5	17.1	16.1	18.9	25.4	26.2	25.5
		MCDBR	28.6	29.2	30.1	32.2	29.6	25.1	18.9	18.1	21.3	26.5	28.5	29.4
		MCWBR	14.7	12.8	10.3	8.8	9.4	8.3	6.5	6.4	7.1	9.4	12.7	15.2
	5% WB	MCDBR	25.4	25.5	27.5	28.5	25.8	20.7	17.8	16.9	18.5	23.5	26.0	26.8
	MCWBR	13.9	12.2	10.5	10.1	9.9	8.4	7.1	6.7	7.2	9.4	12.6	14.8	
Clear-Sky Solar Irradiance	taub	0.484	0.490	0.525	0.599	0.663	0.756	0.811	0.719	0.603	0.577	0.573	0.527	
	taud	1.859	1.887	1.824	1.700	1.608	1.501	1.450	1.595	1.767	1.723	1.693	1.774	
	Ebn at Noon	229	243	244	231	217	197	185	201	221	214	200	207	
	Edh at Noon	55	58	66	77	84	94	98	84	69	68	65	58	
All-Sky Solar Radiation	RadAvg	1163	1467	1866	2138	2275	2186	2024	1952	1880	1664	1295	1123	
	RadStd	71	95	82	61	46	60	79	88	76	68	66	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page